

EXPLAINED EXAMPLE 8 – ISENTROPIC EFFICIENCY

- ① A steam turbine operates with an inlet temperature of 780 K and inlet pressure of 3 MPa. The exit pressure of the turbine is 0.7 MPa and if the mass flow is 450 kg/s, the work output of the turbine is 177.7 MW. Assuming steady, adiabatic operation, what is the final temperature of the fluid and the isentropic efficiency of the turbine?

Given :

- ② $T_1 = 780 \text{ K}$
- $P_1 = 3 \text{ MPa}$
- $P_2 = 0.7 \text{ MPa}$
- $\dot{m} = 450 \text{ kg/s}$
- $\dot{W}_{\text{out}} = 177.7 \text{ MW}$

Find : T_2
: η_{turb}

Assume : S.C.S.
① : Steady-flow
: $\Delta \text{ke} = \Delta \text{pe} = 0$
: $\dot{Q} = 0$

→ ② Energy equation: $\dot{m}(\Delta h + \cancel{\Delta \text{ke}} + \cancel{\Delta \text{pe}}) = \cancel{\dot{Q}} - \dot{W}$
 $\dot{m} \Delta h = -\dot{W}$
 $h_2 = h_1 - \frac{\dot{W}}{\dot{m}} \rightarrow \text{find } T_2 \text{ from } h_2 \text{ \& } P_2$

→ ① State 1 : $P_1 = 3 \text{ MPa}$
 $T_1 = 780 \text{ K}$ } Tabu D.2 → $P_1 = 3 \text{ MPa}$
 $T_1 > T_{\text{sat}} \rightarrow \text{vapor}$

: Table D.3L → $h_1 = 3472.6 \frac{\text{kJ}}{\text{kg}}$
 $s_1 = 7.2557 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$

→ ② State 2 : $h_2 = h_1 - \frac{\dot{W}}{\dot{m}} = 3472.6 \frac{\text{MJ}}{\text{kg}} - \frac{177.7}{450} \frac{\text{MJ}}{\text{kg}}$
 $h_2 = 3077.7 \frac{\text{kJ}}{\text{kg}}$

: Table D.2 → $P_2 = 0.7 \text{ MPa}$
 $h_2 > h_g \rightarrow \text{vapor}$

! Table D.3H $\rightarrow h_2 = 3077.7 \frac{\text{kJ}}{\text{kg}}$

$$\boxed{T_2 \approx 580 \text{ K}} \quad (\text{closest temp})$$

① $\rightarrow \eta_{\text{turb}} = \frac{\dot{W}_{\text{act}}}{\dot{W}_{\text{ideal}}} \rightarrow \dot{W}_{\text{ideal}} = -\dot{m} (h_{2s} - h_1)$

where h_{2s} from isentropic expansion

② $\Rightarrow$ State 2s: $P_2 = 0.7 \text{ MPa}$

$$s_{2s} = s_1 = 7.2557 \frac{\text{kJ}}{\text{kg}}$$

! Table D.2 $\rightarrow s_{2s} > s_g \rightarrow \text{vapor}$

! Table D.3H $\rightarrow h_{2s} \approx 3031.9 \frac{\text{kJ}}{\text{kg}}$

①

$$\begin{aligned} \Rightarrow \dot{W}_{\text{ideal}} &= -\dot{m} (h_{2s} - h_1) \\ &= -450 (3031.9 - 3472.6) \\ &= 198.315 \text{ MW} \end{aligned}$$

②

$$\Rightarrow \eta_{\text{turb}} = \frac{177.7}{198.315} \rightarrow \boxed{\eta_{\text{turb}} = 0.896}$$